

Certification Topics and Weightage

Exam Objectives	Questions Weightage
OCI Data Science Introduction and Configuration	10%
Design and Set Up the OCI Data Science Workspace	15%
Implement the End-to-End Machine Learning Life Cycle	45%
Apply MLOps Practices	20%
Use Related OCI Services	10%
Total	100%

Exam Topics

Exam Objectives	Concepts Evaluated
Implement the End-to-End Machine Learning Life Cycle	- ML Lifecycle overview - Accessing data from different sources - Exploring and preparing data - Visualizing and profiling data - Creating and training models using OCI - Model training with Oracle AutoML - Model evaluation - Model explanations - Managing models using Model Catalog - Deploying and invoking a catalogued model - OCI Generative AI Integration - Application deployment to Data Science - Operators

accessing data from different sources, exploring and preparing data

Prepare for the exam

- Complete Oracle University's *Oracle Cloud Infrastructure Data Science Professional* training course
- Go through the OCI Data Science documentation and take notes
- Go through Oracle Accelerated Data Science documentation
- Go through OCI AI/Data Science blog articles

Which Oracle Cloud Infrastructure (OCI) Data Science policy is invalid?

- A. Allow ~~group~~ DataScienceGroup to use ~~virtual-network-family~~ in compartment DataScience **VALID**
- B. Allow group DataScienceGroup to use data-science-model-sessions in compartment DataScience **ANSWER**
- C. Allow ~~dynamic-group~~ DataScienceDynamicGroup to ~~manage data science projects~~ in compartment DataScience **VALID**
- D. Allow ~~dynamic-group~~ DataScienceDynamicGroup to ~~manage data science family~~ in compartment DataScience **VALID**

data-science-notebook-sessions is a reserved keyword for an individual resource type, but data-science-model-sessions is not
Allow group DataScienceGroup to use data-science-model-sessions in compartment DataScience

Contains your custom logic for loading serialized model objects to memory, and defines an inference endpoint (predict())
score.py
Lists the third-party dependencies that you must install in your local environment before running introspection tests
requirements.txt

Provides an optional series of test definitions that you can run on your model artifact before saving it to the model catalog
model_artifact_validate.py

You are working as a data scientist for a healthcare company. You have analyzed a series of neurophysiological data on OCI Data Science and have developed a convolutional neural network (CNN) classification model. It predicts the source of seizures in drug-resistant epileptic patients.

You created a model artifact with all the necessary files. When you deployed the model, it failed to run because you did not point to the correct conda environment in the model artifact.

Where would you provide instructions to use the correct conda environment?

- A. ~~score.py~~
- B. runtime.yaml **ANSWER**
- C. ~~requirements.txt~~
- D. ~~model_artifact_validate.py~~

Provides instructions about which conda environment to use when deploying the model using a Data Science model deployment
runtime.yaml

1.2 to include 20% buffer

Exam Topics

Exam Objectives	Concepts Evaluated
OCI Data Science Introduction and Configuration	- OCI Data Science overview and concepts - Accelerated Data Science (ADS) SDK overview and capabilities - Tenancy configuration for Data Science
Design and Set Up the OCI Data Science Workspace	- Creating and managing Projects and Notebook sessions - Creating and managing Conda environments - Using OCI Vault in OCI Data Science - Configuring and managing source code in Code Repositories (Git)

Exam Topics

Exam Objectives	Concepts Evaluated
Apply MLOps Practices	- MLOps overview and architecture - OCI Data Science jobs - Scaling in OCI Data Science - Autoscaling Model deployment for Inference - Monitoring and logging - Pipelines to automate machine learning workflow
Use Related OCI Services	- Creating and managing Spark Applications using OCI Data Flow and OCI Data Science - OCI Open Data Service overview - Create and Export a Dataset using OCI Data Labeling

For the fourth exam

The run limit for a notebook cell is about 30 minutes.

Ensure you don't execute long-running Python processes in a notebook cell. Run the process directly in the terminal and use Python logging to get updates on the progress of your job.

Alternatively, you can run multiple notebooks all using the same OCPUs and memory. Avoid having multiple users in the same notebook session due to the possibility of resource contention and write conflicts.

A resource principal enables resources to be authorized *principal* actors with their own identity, and it authenticates using the certificates that are added to it. These certificates are automatically created, assigned to resources, and rotated, avoiding the need for you to store credentials in your notebook session. Authenticate using your notebook session's resource principal...

Q2: Set Up the OCI Data Science Workspace

Which is NOT a valid OCI Data Science notebook session approach?

- A. ~~Ensure you don't execute long-running Python processes in a notebook cell. Run the process directly in the terminal and use Python logging to get updates on the progress of your job.~~ **VALID**
- B. ~~Avoid having multiple users in the same notebook session due to the possibility of resource contention and write conflicts.~~ **VALID**
- C. While connecting to data in OCI Object Storage from your notebook session, the best practice is to make a local copy on the device and then upload it to your notebook session block volume. **ANSWER**
- D. ~~Authenticate using your notebook session's resource principal to access other OCI resources. Resource principals provide a more secure way to authenticate to resources compared to the OCI configuration and API key approach.~~ **VALID**

To retrieve your data, first set up a connection to OCI Object Storage. After this, you can use the OCI Python SDK or ADS SDK in a notebook session to retrieve your data from Object Storage.

While connecting to data in OCI Object Storage from your notebook session...

Q4: Apply MLOps Practices

You have an image classification model in the model catalog which is deployed as an HTTP endpoint using model deployments. Your tenancy administrator is seeing increased demands and has asked you to increase the load balancing bandwidth from the default of 10 Mbps. You are provided with the following information:

Payload size in KB: 1024

Estimated requests per second: 120 requests/second (Monday through Friday, in every month, in every year)

Buffer percentage: 20%

What is the optimal load balancing bandwidth to redeploy your model?

- A. 452 Mbps
- B. 52 Mbps
- C. 7052 Mbps
- D. 1152 Mbps

You can use the following formula to estimate the bandwidth of the load balancer you need. Additionally, keep an extra 20% as a buffer percentage to allow for estimation errors and sporadic peak traffic.
(Payload size in KB) * (Estimated requests per second) * 8 / 1024
Here, payload is 1,024 KB and you estimate 120 requests per second. The recommended load balancer bandwidth for the model deployment would be (1024 * 120 * 8 / 1024) * 1.2 = 1152 Mbps.

Q5: Use Related OCI Services

You want to create an anomaly detection model using the **OCI Anomaly Detection** service that **avoids as many false alarms** as possible. **False Alarm Probability (FAP)** indicates model performance.

How would you **set the value** of False Alarm Probability?

- A. High
- B. Low
- C. Zero
- D. Use a function

Lower FAP indicates better model performance.

Lower FAP indicates better model performance. Setting it low as per model performance is a better choice than always setting it to 0.

✔ Oracle Resource Manager (ORM) automatically creates the policies for a basic use case, required user groups, and dynamic groups. It doesn't create individual Data Science users.

1. What does the Data Science Service template in Oracle Resource Manager (ORM) NOT automatically create?

- ☐ Dynamic groups
- ☒ Individual Data Science users ✔
- ☐ Policies for a basic use case
- ☐ Required user groups

✘ You have chosen an incorrect answer.

2. Which OCI Data Science interaction method can function in both ways — with or without the need of scripting?

- ☐ OCI Console ✔
- ☒ REST API ✘
- ☐ CLI
- ☐ Language SDKs

✔ The model catalog is a centralized and managed repository of model artifacts. A stored model includes metadata about the provenance of the model including Git-related information and the script or notebook used to push the model to the catalog. Models stored in the model catalog can be shared across members of a team and they can be loaded back into a Notebook session.

3. Which allows the sharing and loading back of ML models into a Notebook session?

- ☐ Model provenance
- ☐ Model taxonomy
- ☒ Model catalog ✔
- ☐ Model deployment

✘ You have chosen an incorrect answer.

4. Which statement about dynamic groups is true?

- ☒ They are a logical grouping of resources that can be accessed only by certain groups that have received administrator permission. ✘
- ☐ They define what Data Science principals, such as users and resources, have access to in OCI.
- ☐ They define a set of matching rules that determine group membership instead of requiring members to be added explicitly. ✔
- ☐ They are individual users that are grouped in OCI by administrators and granted access to Data Science resources within compartments.

✔ A resource principal can be used to authenticate and access Oracle Cloud Infrastructure (OCI) resources. The resource principal consists of a temporary session token and secure credentials that enable the database to authenticate itself to other OCI services. To use the resource principal, you or your tenancy administrator define the OCI policies and a dynamic group that allows you to access OCI resources with a resource principal.

5. Which statement about resource principals is true?

- ☐ A resource principal is not a secure way to authenticate to resources, compared to the OCI configuration and API key approach.
- ☐ The Data Science service does not provide authentication via a notebook session's or job run's resource principal to access other OCI resources.
- ☐ When you authenticate using a resource principal, you need to create and manage credentials to access OCI resources.
- ☒ A resource principal is a feature of IAM that enables resources to be authorized principal actors. ✔

✔ OCI Notebook sessions are interactive coding environments for building and training models. A Notebook session is created inside a Data Science project.

6. Which feature of Oracle Cloud Infrastructure Data Science provides an interactive coding environment for building and training machine learning models?

- ☐ Model catalog
- ☒ Notebook session ✔
- ☐ Projects
- ☐ Jobs

✔ Feature engineering is used to represent the data in a way that helps the model learn more effectively.

7. What is feature engineering in machine learning used for?

- ☐ To help understand the data set features
- ☒ To represent the data in a way that helps the model learn more effectively. ✔
- ☐ To perform parameter tuning
- ☐ To interpret ML models

- ✓ The process of building and breaking the ML model has six steps. Of these, Data Exploration is used to cleanse the data by identifying corrupt, inaccurate, and incomplete data and replacing or deleting it. This also helps in identifying the imbalances present in the data.

1. Which stage in the machine learning life cycle helps identify imbalances present in data?

- ☒ Data Exploration ✓
- ☐ Data Monitoring
- ☐ Data Modeling
- ☐ Data Access

- ✓ This file contains the function definitions that are necessary to load your model to memory and make predictions. The two functions are called `load_model()` and `predict()`. The function parameters are not customizable. For example, you can define data transformations in `predict()` before calling the inference method of your estimator object. In other words, it is used for executing the inference logic code.

2. What do you use the `score.py` file for?

- ☐ Defining the required Conda environments
- ☐ Defining the scaling strategy
- ☐ Configuring the deployment infrastructure
- ☒ Executing the inference logic code ✓

- ✓ Anomaly detection is used for finding anomalies or any irregularities in your data.

3. You are working for a bank. You are required to analyze customer accounts' access data and flag any irregular access attempts. Which OCI Data Science operator are you most likely to use?

- ☐ PII
- ☐ Forecasting
- ☒ Anomaly ✓

- ✓ The machine learning life cycle has six important tasks: Data Access, Data Exploration and Data Preparation, Modeling, Validation, Model Deployment, and Monitoring/Refresh/Retirement. Database Management is not a part of this life cycle.

4. Which activity is NOT a part of the machine learning life cycle?

- ☐ Modeling
- ☐ Data Access
- ☒ Database Management ✓
- ☐ Model Deployment

- ✓ Oracle AutoML automates four major steps in the machine learning modeling process to deliver significant productivity improvements for data scientists. They are Algorithm Selection, Adaptive Sampling, Feature Selection, and Model Tuning.

5. Which step is a part of AutoML pipeline?

- ☒ Feature Selection ✓
- ☐ Model Saved to Model Catalog
- ☐ Feature Extraction
- ☐ Model Deployment

- ✓ The objective of the loss function is to represent the difference between the predicted values $\hat{Y}$ and the true targets Y as a single number, the loss score. The loss function is also known as the cost function.

6. Which function represents the difference between the predictive value and the target value?

- ☐ Fit function
- ☒ Cost function ✓
- ☐ Optimizer function
- ☐ Update function

✔ Machine learning models are data driven. Whenever the data changes, the model needs retraining. Else the performance degrades over time.

1. Which statement is true about machine learning models?

- ☐ Static predictions become increasingly accurate over time.
- ☒ Model performance degrades over time due to changes in data. ✔
- ☐ Data models are more static and generally require fewer updates than software code.
- ☐ A high-quality model will not need to be retrained as new information is received.

✔ A job is a template that describes the task. In each job run, you can override some of the job configuration, and most importantly, the environment variables and CLI arguments. You can have the same job with several sequentially or simultaneously started job runs with different parameters.

2. Which statement is true about Oracle Cloud Infrastructure Data Science Jobs?

- ☒ Jobs provisions the infrastructure to run a process on demand. ✔
- ☐ Jobs comes with a set of standard tasks that cannot be customized.
- ☐ You must create and manage your own Jobs infrastructure.
- ☐ You must use a single Shell/Bash or Python artifact to run a job.

✘ You have chosen an incorrect answer.

3. Why would you use a mini batch when processing a job in Data Science Jobs?

- ☐ You want to process data frequently. ✔
- ☐ There is a small amount of total data to process.
- ☐ You do not need to process data quickly.
- ☒ You want several distributed models to run simultaneously. ✘

✔ Action is a user-defined response to an event, such as triggering a notification. Triggering a Pager Duty notification in this case is an example of Action.

4. In the OCI Console, as part of Monitoring, of what is triggering a Pager Duty notification an example?

- ☐ Function
- ☐ Event
- ☐ Rule
- ☒ Action ✔

✔ When working with custom logs, the customer captures log events collected in a particular context. Here the standard output (stdout) and standard error (stderr) outputs from job artifacts are automatically captured.

5. Which statement is true about logs for Oracle Cloud Infrastructure Jobs?

- ☐ Logs are automatically deleted when the job and job run are deleted.
- ☒ All stdout and stderr are automatically stored when automatic log creation is enabled. ✔
- ☐ Integrating Data Science Jobs resources with Logging is mandatory.
- ☐ Each job run sends outputs to a single log for that job.

✔ Continuous Training is unique to MLOps and refers to the automatic retraining of ML models for redeployment.

6. Which step is unique to MLOps, as opposed to DevOps?

- ☐ Continuous Delivery
- ☐ Continuous Integration
- ☐ Continuous Deployment
- ☒ Continuous Training ✔

✔ You can enable autoscaling for an existing model deployment or update any existing autoscaling configuration. For model deployments in an Active state, note that modifications to the Autoscaling Scaling Policy fields must occur independently, without simultaneous changes to other configurations. However, updates to fields such as Display Name, Description, Tags, and other non-infrastructure-related aspects are allowed. In contrast, when the model deployment is Inactive, you have the flexibility to change all options simultaneously.

7. You want to change the autoscaling configuration for infrastructure and non-infrastructure fields for an existing model deployment in an Active state in Oracle Data Science. Which statement is true?

- ☐ You can modify the Autoscaling Scaling Policy fields and other configurations simultaneously regardless of its state.
- ☐ Infrastructure-related aspects can't be modified for a model deployment, regardless of state.
- ☒ You must disable the model deployment to update the Autoscaling Scaling Policy fields and other configurations. ✔
- ☐ Non-infrastructure-related aspects can't be modified for an active model deployment.

✘ You have chosen an incorrect answer.

8. As a data scientist, you require a pipeline to train ML models. When can a pipeline run be initiated?

- ☐ After the Active state ✔
- ☐ During the pipeline run state
- ☐ After it is created
- ☒ Before the Active state ✘

✔ Oracle Open Data is a service from Oracle for Research that makes large data sets available to researchers around the world. It curates information like spatial images, protein sequences, and annotated text files from the world's leading scientific data sources.

1. You are a researcher who requires access to large data sets. Which OCI service would you use?

- ☐ Oracle Databases
- ☐ OCI Data Science
- ☒ Oracle Open Data ✔
- ☐ ADW

✔ PySpark 3.2 and the Big Data service leverage the power of Apache Spark and MLlib to speed up model building.

2. What is the name of the machine learning library used in Apache Spark?

- ☒ MLlib ✔
- ☐ Structured Streaming
- ☐ GraphX
- ☐ HadoopML

✘ You have chosen an incorrect answer.

3. Which option indicates the three types of data used for Data Labeling?

- ☐ Graphic, text, document
- ☐ Image, text, document ✔
- ☒ Image, audio, document ✘
- ☐ Text, audio, video

✘ You have chosen an incorrect answer.

4. Which OCI service provides a scalable environment for developers and data scientists to run Apache Spark applications at scale?

- ☐ Anomaly Detection
- ☒ Data Science ✘
- ☐ Data Labeling
- ☐ Data Flow ✔

✔ Cramer's V method is specifically designed for measuring the association between two nominal (categorical) variables. Unlike Pearson's correlation, which measures linear relationships between continuous variables, Cramer's V captures the strength of association in categorical data.

1. Which correlation method is used to measure the relationship between two categorical variables in ADS?

- ☐ Pearson correlation coefficient
- ☐ Spearman correlation coefficient
- ☐ Chi-square test
- ☒ Cramer's V method ✔

✔ Infrastructure is automatically deprovisioned when a job run completes or is canceled, ensuring cost efficiency.

2. A company is running a job in OCI Data Science Jobs and wants to ensure that the infrastructure is deprovisioned immediately after the job completes to avoid unnecessary costs. What happens when the job ends?

- ☐ The infrastructure remains active for 30 days.
- ☐ The compute shape is reset to default.
- ☒ The infrastructure is automatically deprovisioned. ✔
- ☐ The job artifact is deleted.

✔ PII Operator is used to redact the sensitive information.

3. You are given data with sensitive information like emails and contact numbers of the customer. You need to redact this information before using it further. What OCI Data Science Operator can you use for this purpose?

- ☐ Forecasting
- ☐ Anomaly
- ☒ PII ✔

✔ With AI Quick Actions , you can fine tune the model with custom data and deploy it.

4. You have custom data and you want to customize an off-the-shelf LLM and deploy it quickly. How can AI Quick Actions help you?

- ☐ To deploy the of-the-shelf model
- ☒ To fine-tune the model and deploy ✔
- ☐ To pretrain the LLM

✔ The correct sequence is , Load documents, Split documents, Embed documents, Create vector database from documents, Create retriever, Create chain, Create model, Prepare model artifacts, Verify model, Save model, Deploy model.

5. When deploying an RAG application to OCI Data Science, what is the correct sequence of steps you would need to follow?

- ☒
 - 1. Load documents.
 - 2. Split documents.
 - 3. Embed documents.
 - 4. Create vector database from documents.
 - 5. Create retriever.
 - 6. Create chain.
 - 7. Create model.
 - 8. Prepare model artifacts.
 - 9. Verify model.
 - 10. Save model.
 - 11. Deploy model.
- ☐
 - 1. Create model.
 - 2. Load documents.
 - 3. Split documents.
 - 4. Embed documents.
 - 5. Create vector database from documents.
 - 6. Create retriever.
 - 7. Create chain.
 - 8. Prepare model artifacts.
 - 9. Verify model.
 - 10. Save model.
 - 11. Deploy model.
- ☐
 - 1. Load documents.
 - 2. Split documents.
 - 3. Embed documents.
 - 4. Create vector database from documents.
 - 5. Prepare model artifacts.
 - 6. Create retriever.
 - 7. Create chain.
 - 8. Create model.
 - 9. Verify model.
 - 10. Save model.
 - 11. Deploy model.

✔ PDP shows the average effect of a feature on model predictions across all data points, while ICE examines how each individual sample's prediction changes when a feature value is altered. ICE provides more granular insights, capturing heterogeneity among different samples.

6. What is the key difference between PDP (Partial Dependence Plot) and ICE (Individual Conditional Expectation) in ADS?

- ☐ PDP is used for classification, while ICE is only used for regression.
- ☐ PDP is a supervised learning technique, while ICE is used for unsupervised learning.
- ☒ PDP provides feature-level insights, while ICE provides sample-level insights. ✔
- ☐ PDP works only for categorical features, while ICE works only for continuous features.

✔ Data Science notebook session as an Interactive Development Environment (IDE) is used for iteratively developing PySpark applications.

7. A data scientist wants to develop a PySpark application iteratively using a sample of their dataset. Which environment is recommended for this purpose?

- ☐ OCI Compute
- ☐ OCI Object Storage
- ☐ OCI Virtual Cloud Network
- ☒ OCI Data Science notebook session ✔

✔ Predefined autoscaling policies can use metrics like CPU utilization or memory utilization.

8. A team wants to use CPU utilization as a metric to trigger autoscaling. Which type of autoscaling policy should they configure?

- ☐ Custom scaling metric
- ☒ Predefined metric ✔
- ☐ Manual scaling
- ☐ Load balancer scaling

✔ Conda environments allow data scientists to create isolated software setups, preventing conflicts between different libraries and dependencies used for various models, such as computer vision and regression models.

9. What is the primary advantage of using Conda environments in Data Science?

- ☐ They help in compressing datasets for storage efficiency.
- ☐ They provide faster GPU processing speeds.
- ☐ They replace the need for cloud storage in machine learning projects.
- ☒ They enable isolated software configurations for different projects. ✔

✔ Autoscaling dynamically adjusts compute resources based on demand, meaning that cost fluctuates with resource utilization. If more instances are needed during high traffic, costs increase, and if fewer resources are needed, costs decrease.

10. Which statement is incorrect regarding the benefits of autoscaling for model deployment in Oracle Data Science?

- ☒ By using autoscaling, the cost of deployment remains constant irrespective of resource utilization. ✔
- ☐ Autoscaling, in conjunction with load balancers, enhances availability by rerouting traffic to healthy instances in case of instance failure.
- ☐ Autoscaling dynamically adjusts compute resources based on real-time demand, ensuring efficient handling of varying loads.
- ☐ Users can set customizable triggers for autoscaling using MQL expressions to tailor the scaling behavior according to specific needs.

Dynamic groups contain resource principals, such as notebook sessions or job runs, that can make API calls based on defined policies. Unlike user groups, dynamic groups are based on resource attributes rather than user identities.

11. What is the purpose of a dynamic group in OCI?

- ☐ To allocate computing resources dynamically
- ☐ To group individual users for easier management
- ☒ To manage API access for resources such as notebook sessions
- ☐ To define storage limits for data science resources

When a model deployment is deactivated, its associated VM instances and load balancer shut down, stopping billing. However, the model's metadata remains saved, allowing it to be reactivated later with the same endpoint.

12. What happens when a model deployment in OCI Data Science is deactivated?

- ☐ The model deployment metadata is erased along with the model artifacts.
- ☐ The deployed model is permanently deleted, and predictions are no longer possible.
- ☒ The model's HTTP endpoint becomes unavailable, but metadata is preserved.
- ☐ The model remains active but stops accepting new inference requests.

Pushing changes to a remote repository involves transferring a significant amount of data over the internet, which can be slow if the internet speed is low. Similarly, updating the local repository to match the content from a remote repository also involves downloading a large amount of data, which can be time-consuming if the internet speed is slow. All other options are local operations and does not involve any network activity.

13. While working with Git on Oracle Cloud Infrastructure (OCI) Data Science, you notice that two of the operations are taking more time than the others due to your slow internet speed. Which two operations would experience the delay?

- ☒ Pushing changes to a remote repository
- ☒ Converting an existing local project folder to Git repository
- ☐ Moving changes into the staging area for the next commit
- ☐ Updating the local repo to match the content from a remote repository
- ☐ Making a commit that is taking a snapshot of the local repository for the next push

Custom scaling metrics allow users to create sophisticated queries using NQL expressions, including logical operators like AND/OR.

14. A team wants to create a sophisticated autoscaling query that combines multiple metrics using logical operators. Which option should they use?

- ☐ Load balancer scaling
- ☐ Cooldown periods
- ☒ Custom scaling metric with NQL expressions
- ☐ Predefined metrics

The Model Catalog allows the scientist to immutably store and track the model, ensuring reproducibility and auditability. This ensures that the model can be retrieved in the exact state it was in when saved.

15. A data scientist is working on a fraud detection model. They need to store the trained model so that it can be versioned, tracked, and later deployed without modification.

Which feature should they use?

- ☐ Model Deployment
- ☒ Model Catalog
- ☐ Hyperparameter Tuning
- ☐ Model Explainability

When a notebook session is deactivated, block storage remains, allowing data and results to be accessed after reactivation. However, any data stored on the boot volume is lost.

16. A data scientist is running a long-term experiment in an OCI notebook session. They need to save results even if they deactivate the session to reduce costs.

What should they do?

- ☐ Keep the session active indefinitely to prevent data loss.
- ☐ Use default networking to automatically back up results to OCI Object Storage.
- ☒ Store all results in the block storage, as it persists after deactivation.
- ☐ Save results only in the boot volume, as it is retained indefinitely.

The loss function calculates the difference between the predicted values and the actual target values, representing this difference as a single numerical loss score. The lower the loss score, the better the model's predictions match the true values.

17. What is the primary goal of the loss function in model training?

- ☐ To determine the best algorithm for training the model
- ☒ To compare predicted values with true target values and quantify their difference
- ☐ To maximize the likelihood of data points fitting the model
- ☐ To update the model parameters to optimize performance

Storing credentials in OCI Vault ensures secure access control and prevents leaks, as credentials are retrieved only when needed and not stored in code or configuration files.

18. A data scientist needs to securely access an external database from their notebook session.

What is the best way to store the credentials?

- ☐ Share the credentials via email with team members.
- ☐ Store the credentials in a plaintext configuration file.
- ☐ Hardcode the credentials in the Jupyter Notebook.
- ☒ Save the credentials in OCI Vault and retrieve them programmatically when needed.

Oracle Cloud Infrastructure (OCI) Data Labeling supports annotating Document, Image and Text labeling.

19. You have been given a collection of digital files required for a business audit. They consist of several different formats that you would like to annotate using Oracle Cloud Infrastructure (OCI) Data Labeling.

Which three types of files could this tool annotate?

- ☐ An audio recording of a phone conversation
- ☒ A type-written document that details an annual budget
- ☐ Video footage of a conversation in a conference room
- ☒ Images of computer server racks
- ☒ A collection of purchase orders for office supplies

In Oracle Data Science, modifications to an active model deployment's Autoscaling Scaling Policy require changes to be made one field at a time. This ensures system stability and prevents conflicting updates. Other configurations cannot be updated simultaneously while modifying autoscaling parameters.

20. Which statement is true regarding autoscaling configuration for an existing model deployment in an Active state in Oracle Data Science?

- ☐ Changes to the Autoscaling Scaling Policy fields must occur one field at a time, without simultaneous changes to other configurations.
- ☐ Only non-infrastructure-related aspects can be modified for an active model deployment.
- ☒ You must disable the model deployment to update the Autoscaling Scaling Policy fields.
- ☐ You can modify the Autoscaling Scaling Policy fields and other configurations simultaneously.

Since the regulator is asking about a specific prediction, LIME is the best technique. It provides a local explanation using an interpretable surrogate model to explain why a particular transaction was classified as fraudulent.

21. A data scientist is using an AI model to predict fraudulent transactions. A financial regulator asks why a specific transaction was flagged as fraud.

Which technique should the data scientist use?

- ☐ What-If Explanation
- ☐ Global Explanation
- ☒ Local Explanation
- ☐ Feature Permutation Importance

"Model taxonomy allows you to describe the model you are saving to the model catalog. You can use preset fields to document the: Machine learning use case Machine learning model framework Version Estimator object Hyperparameters Artifact test results"

22. You have just started as a data scientist at a healthcare company. You have been asked to analyze and improve a deep neural network model that was built based on the electrocardiogram records of patients. There are no details about the model framework that was built.

What would be the best way to find more details about the machine learning models inside model catalog?

- ☐ Check for Provenance details.
- ☐ Check for metadata tags
- ☒ Check for Model Taxonomy details.
- ☐ Refer to the code inside the model.

Oracle Cloud Infrastructure Data Science Jobs is a service that enables data scientists to define and run repeatable tasks, such as data processing or model training, on fully managed infrastructure. Jobs provides a platform for automating data science workflows and simplifying the management of compute resources.

23. Which statement best describes Oracle Cloud Infrastructure Data Science Jobs?

- ☒ Jobs lets you define and run repeatable tasks on fully-managed infrastructure.
- ☐ Jobs lets you define and run all Oracle Cloud DevOps workloads.
- ☐ Jobs lets you define and run repeatable tasks on customer-managed infrastructure.
- ☐ Jobs lets you define and run repeatable tasks on fully-managed, third-party cloud infrastructure.

Geospatial data, protein sequences, genomic data, and annotated text files are examples of data available in Oracle Open Data. Financial transaction data is not supported.

24. Which type of data is NOT available in Oracle Open Data?

- ☐ Annotated text files
- ☐ Geospatial data from satellite systems
- ☐ Protein sequences and genomic data
- ☒ Financial transaction data

AI Quick Actions provide a fast and easy way to deploy pre-trained LLMs without requiring extensive coding or configuration, making it the ideal choice for quickly integrating a model into a chatbot.

25. A company wants to integrate an LLM into its customer support chatbot using OCI. What is the fastest way to deploy and test the model?

- ☐ Manually configuring a model deployment using OCI SDK
- ☐ Building a deep learning model in Jupyter Notebook
- ☐ Training a custom model from scratch using OCI AutoML
- ☒ Using AI Quick Actions to quickly deploy a pretrained LLM

The PII Detection Operator helps detect and redact personal identifiable information (PII) from datasets, ensuring data privacy compliance before sharing sensitive information.

26. A healthcare company needs to redact personal details (such as names, emails, and phone numbers) from patient records before sharing them with a research institute.

Which operator is best suited for this task?

- ☒ PII Detection Operator
- ☐ Clustering Operator
- ☐ Anomaly Detection Operator
- ☐ Forecasting Operator

"Read" means the users will be able to access and view the details of data science resources, but they won't have the permission to modify, delete or create them. This verb is commonly used in access control policies to grant read-only access to users or groups.

27. You want to create a user group for a team of external data science consultants. The consultants should only have the ability to view data science resource details but not the ability to create, delete, or update data science resources.

What verb should you write in the policy?

- ☒ Read
- ☐ Manage
- ☐ Use
- ☐ Inspect

Users can create a Data Science Project via the OCI Console UI (manual selection) or programmatically using the ADS SDK with Python to automate project creation.

28. A team of data scientists is working on multiple machine learning models for fraud detection. They want to collaborate in a structured manner.

What option is available to create a Data Science Project in OCI?

- ☒ Can be created through either the OCI Console UI or the ADS SDK
- ☐ Can be created using a command-line interface (CLI) only
- ☐ Can be created only through the OCI Console UI
- ☐ Can be created only through the ADS SDK

Machine learning Lifecycle order :i.e. Data Access, Data Exploration, Feature Exploration, Feature Engineering, Modeling.

29. A bike sharing platform has collected user commute data for the past three years. For increasing the profitability and making useful inferences, a machine learning model needs to be built from the accumulated data.

Which option has the correct order of the required machine learning tasks for building a model?

- ☐ Data Access, Feature Exploration, Data Exploration, Feature Engineering, Modeling
- ☐ Data Access, Feature Exploration, Feature Engineering, Data Exploration, Modeling
- ☒ Data Access, Data Exploration, Feature Engineering, Feature Exploration, Modeling
- ☐ Data Access, Data Exploration, Feature Exploration, Feature Engineering, Modeling

Since resource principal tokens are cached for 15 minutes, any IAM policy or dynamic group changes take time to reflect. The data scientist must wait 15 minutes before the changes take effect.

30. A data scientist updates an IAM policy to grant their notebook session access to an Object Storage bucket. However, the notebook still cannot access the bucket.

What is the likely reason?

- ☒ The resource principal token is still cached.
- ☐ Object Storage does not support access from notebooks.
- ☐ The IAM policy is incorrect.
- ☐ The user needs to restart the entire OCI environment.

For users new to forecasting, the Operator also has an auto-select option. This is the most computationally expensive option as it splits the training data into several validation sets, evaluates each framework, and tries to decide the best one.

31. What model parameter value are you most likely to use if you are not sure of your selection while configuring the Forecasting operator?

- ☐ autots
- ☐ arima
- ☒ auto
- ☐ prophet

Dependencies can be set to create a logical workflow, ensuring steps are executed in the desired sequence.

32. How can a team ensure that data processing occurs before model training in a pipeline?

- ☒ By setting dependencies between steps
- ☐ By using the same programming language for all steps
- ☐ By increasing the block volume size
- ☐ By overriding the default configuration

A custom search space allows users to specify exact hyperparameter ranges based on prior experience, making the tuning process more efficient and targeted.

33. A company has trained a machine learning model and wants to fine-tune it by experimenting with hyperparameter values based on prior experience. What approach should they take?

- ☒ Define a custom search space with specific hyperparameter values.
- ☐ Apply the default search space for broader tuning.
- ☐ Skip hyperparameter tuning altogether.
- ☐ Use the built-in perfunctory search strategy.

Pipeline defaults can be overridden before starting the pipeline execution.

34. You are running a pipeline in OCI Data Science Service and want to override some of the pipeline's default settings. Which statement is true about overriding pipeline defaults?

- ☐ Pipeline defaults can be overridden only by an administrator.
- ☒ Pipeline defaults can be overridden before starting the pipeline execution.
- ☐ Pipeline defaults can be overridden only during pipeline creation.
- ☐ Pipeline defaults cannot be overridden once the pipeline has been created.

To ensure that the 'Age' and 'Education' features are part of the final model, you should set `min_features` equal to a list containing these feature names. In this case, the correct way to define the `min_features` argument would be `min_features = ['Age', 'Education']`. The other options are range, fractions and logical AND operator.

35. You are a data scientist working on census dataset. You have decided to use Oracle AutoML Pipeline for automating your machine learning task and want to ensure that two of the features ("Age" and "Education") are part of the final model that the AutoML creates. To ensure these features are not dropped during the feature selection phase, what would be the best way to define the `min_features` argument in your code?

- ☐ `0 < min_features <= 0.9`
- ☒ `min_features = ['Age', 'Education']`
- ☐ `min_features = 'Age' && min_features = 'Education'`
- ☐ `0 < min_features <= 2`

A job is a template describing the task, while a job run is a single execution of that job, often with customized parameters.

36. What is the difference between a job and a job run in OCI Data Science Jobs?

- ☒ A job is a template, while a job run is a single execution of that template.
- ☐ A job is a single execution, while a job run is a template.
- ☐ A job is immutable, while a job run can be modified.
- ☐ A job is used for model training, while a job run is used for batch inference.

The matching rules in the Data Science Service template define dynamic groups for notebook sessions, model deployments, and job runs, allowing these resources to interact with OCI services securely.

37. Which resource types are included in the default matching rules of the Data Science Service template?

- ☒ `datasciencemodeldeployment`, `datasciencenotebooksession`, `datasciencejobrun`
- ☐ `datascienceanalytics`, `datasciencemonitoring`, `datasciencebatchjob`
- ☐ `datascienceobjectstorage`, `datasciencecomputeinstance`, `datasciencemodeltraining`
- ☐ `datasciencenetwork`, `datasciencedatabase`, `datasciencebackup`

Oracle Cloud Infrastructure Open Data provides access to a variety of curated, high-quality data sets, which are otherwise not available to the public at no cost. Users can subscribe to the Open Data service to gain access to these data sets, which can be used for a variety of purposes, such as research, development, or machine learning.

38. Which two statements are true about Oracle Cloud Infrastructure (OCI) Open Data Service?

- ☒ Open Data is a dataset repository made for the people that create, use, and manipulate datasets.
- ☒ Each dataset in Open Data consists of code and tooling usage examples for consumption and reproducibility.
- ☐ Subscribers can pay and log into Open Data to view curated data sets that are otherwise not available to the public.
- ☐ A primary goal of Open Data is for users to contribute to the data repositories in order to expand the content offered.
- ☐ Open Data includes text and image data repositories for AI and ML. Audio and video formats are not available.

Data labeling is the process of annotating data with properties or labels, which is necessary before training a machine learning model.

39. A data scientist is working on a project to train a machine learning model to identify tigers in images. What is the first step they need to take before training the model?

- ☐ Analyze customer feedback.
- ☐ Deploy the model.
- ☒ Label the images with "tiger" or "not tiger".
- ☐ Use OCI Vision Services.

Feature scaling is important for models like regression and distance-based algorithms (e.g., kNN, SVM, clustering) where different feature scales can negatively impact learning. Normalization (min-max scaling) and standardization (variance scaling) help standardize numerical data for better model performance.

40. What is the primary reason for performing feature scaling in machine learning models?

- ☒ To bring features on to the same scale.
- ☐ To convert categorical data into numerical form.
- ☐ To automatically detect missing values and fill them with mean or median.
- ☐ To make the dataset smaller for faster computation.

After running the Resource Manager stack, the user group is created automatically, but you must manually add users to this group to enable access to Data Science resources.

41. What is the final step after running the Oracle Resource Manager stack for Data Science configuration?

- ☐ Creating an additional stack for security configuration
- ☐ Deleting the default compartment
- ☒ Adding users to the automatically created user group
- ☐ Modifying the Terraform script in GitHub

Prevaling metrics for evaluating multiclass classification model. Check ADS.

42. You are a data scientist using Oracle AutoML to produce a model and you are evaluating the score metric for the model.

Which two prevailing metrics would you use for evaluating the multiclass classification model?

- ☐ Explained variance score
- ☐ Mean squared error
- ☒ Recall
- ☐ R-squared
- ☒ F1 score

Option a) is the correct sequence of steps because it follows a logical order, starting with preparing the model and ending with making predictions on new data.

43. As a data scientist for a hardware company, you have been asked to predict the revenue demand for the upcoming quarter. You develop a time series forecasting model to analyze the data.

Which is the correct sequence of steps to predict the revenue demand values for the upcoming quarter?

- ☐ Verify, prepare model, deploy, save.
- ☒ Prepare model, verify, save, deploy, predict.
- ☐ Prepare model, deploy, verify, save, predict.
- ☐ Predict, deploy, save, verify, prepare model.

The Oracle Data Science service provides a model catalog to store and share machine learning models. The model artifacts that are available for sharing from the model catalog include the trained model file, its metadata, hyperparameters, and evaluation metrics.

44. You have trained a binary classifier for a loan application and saved this model into the model catalog. A colleague wants to examine the model, and you need to share the model with your colleague.

From the model catalog, which model artifacts can be shared?

- ☐ Model metadata and hyperparameters only
- ☐ Metadata, hyperparameters, and metrics only
- ☒ Models, model metadata, hyperparameters, and metrics
- ☐ Models and metrics only

With AI Quick Actions, the deployed model can be invoked using API and CLI.

45. Once you deploy the LLM using AI Quick Actions, how can you invoke your model?

- ☐ Through CLI
- ☒ Through API and CLI
- ☐ Through only CLI
- ☐ Through API

ADS allows database connections without a wallet file by specifying the hostname and port number in connection_parameters. Other options like using HTTP requests or ads.read_sql without credentials won't work.

46. A user wants to fetch data from an Autonomous Database in OCI without using a database wallet. What must they do?

- ☐ Enable API authentication in the database console.
- ☐ Use an HTTP request to retrieve database records.
- ☒ Provide the hostname and port number in the connection_parameters dictionary.
- ☐ Use ads.read_sql without any additional parameters.

After the fine-tuning process, the checkpoints and final outputs are automatically stored in an OCI Object Storage bucket, ensuring persistence and accessibility.

47. Where are the training job outputs stored after fine-tuning is completed?

- ☐ Directly in the OCI Model Catalog
- ☐ In the local storage of the training instance
- ☒ In an OCI Object Storage bucket
- ☐ In a temporary cache that is cleared after job completion

The correct order of the Git repository workflow is: install, configure, and authenticate Git. This step involves installing Git on your local machine, configuring it with your username and email address, and authenticating with the Git hosting service (e.g., GitHub, GitLab, Bitbucket). Configure an SSH key pair and adding the public key to your Git hosting service account to enable secure communication between your local machine and the remote Git repository. Create a local and remote Git repository. This step involves creating a Git repository on your local machine and a remote repository on the Git hosting service, then linking them using the appropriate Git commands. Commit files to the local Git repository. This step involves adding files to the staging area and committing them to the local Git repository using the "git commit" command. Push the commit to the remote Git repository. This step involves pushing the committed changes from the local Git repository to the remote Git repository using the "git push" command. Therefore, the correct order for the Git repository workflow is a) 1, 2, 3, 4, 5."

48. Arrange the following points in the correct Git Repository workflow order.

- 1 - Install, configure, and authenticate Git.
- 2 - Configure SSH keys for the Git repository.
- 3 - Create a local and remote Git repository.
- 4 - Commit files to the local Git repo.
- 5 - Push the commit to the remote Git repo.

- ☐ 3, 5, 1, 2, 4
- ☐ 2, 3, 1, 4, 5
- ☒ 1, 2, 3, 4, 5
- ☐ 4, 2, 3, 1, 5

Automation in MLOps is triggered by changes in data, monitoring events, or calendar intervals, which initiate the retraining and deployment of models.

49. What triggers the automation of the MLOps pipeline?

- ☐ User feedback
- ☒ Changes in data, monitoring events, or calendar intervals
- ☐ Manual intervention by data scientists
- ☐ Random system updates

OCI Vault supports AES, RSA, and ECDSA encryption algorithms. However, SHA-256 is a hashing algorithm, not an encryption algorithm, and is used for data integrity verification rather than encryption.

50. Which is NOT a supported encryption algorithm in OCI Vault?

- ☐ ECDSA (Elliptic Curve Digital Signature Algorithm)
- ☐ AES (Advanced Encryption Standard)
- ☒ SHA-256 (Secure Hash Algorithm 256-bit)
- ☐ RSA (Rivest-Shamir-Adleman)

